

Networks (NW)

09b Fundamentals of Data Transmission

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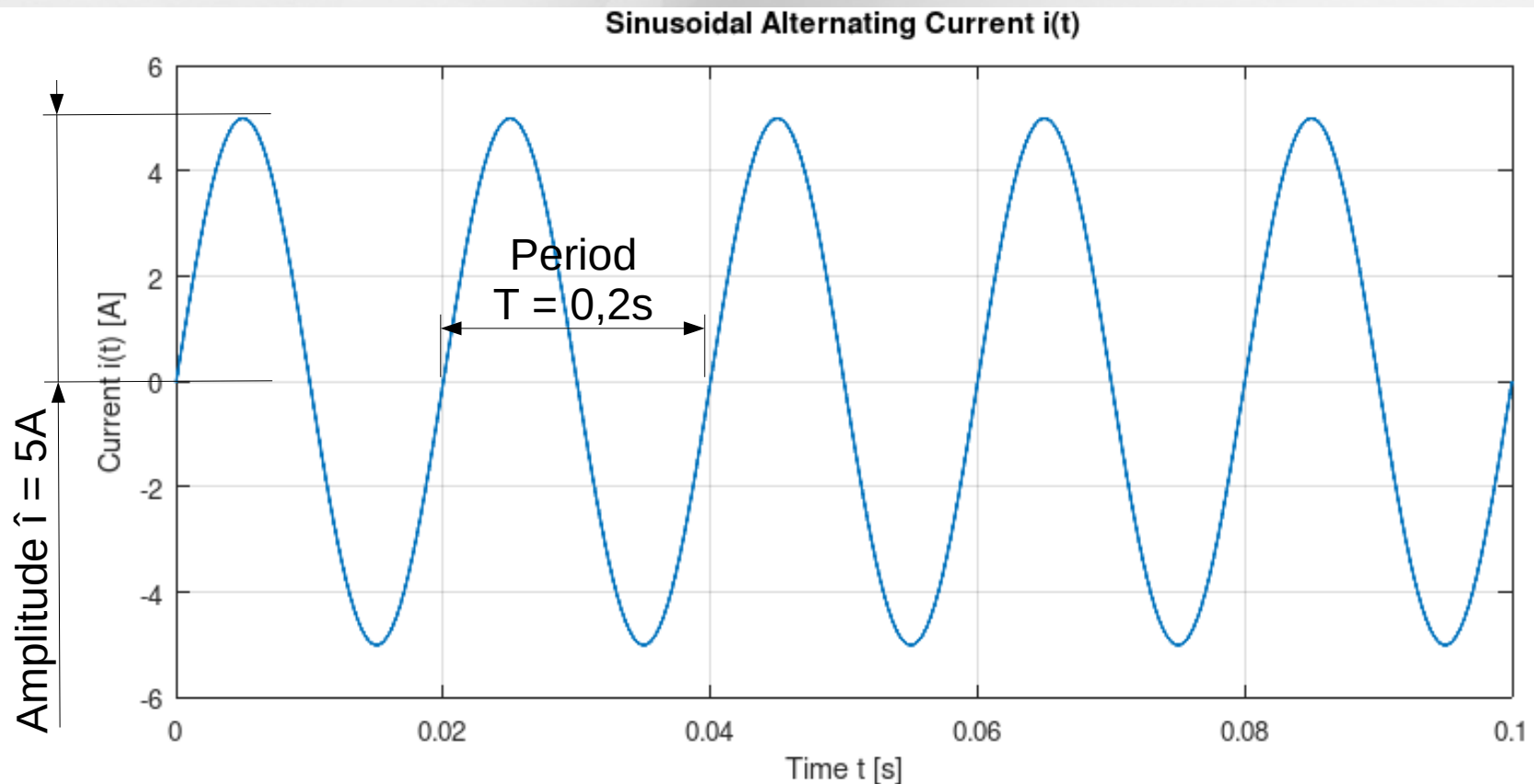
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Basic Concepts of Data Transmission

Electrical Basics

- AC – Alternating Current $i(t)$, Voltage $u(t)$ change periodically
- DC – Direct Current $I = \text{const.}$, Voltage $U = \text{const.}$
- Frequenz f – Number of oscillations per second $f = 1/T$ with $[f] = 1/s = 1 \text{ Hz (Hertz*)}$
 - *Heinrich Hertz, German Physicist, 1857 - 1894
- Period T – Time duration it takes for a complete cycle of the waveform to occur, $T = 1/f$ with $[T] = 1s$
- $f = 1\text{Hz} \rightarrow 1 \text{ oscillation per Second}$
- other $f_{\text{net}} = 50\text{Hz}$, $f_{\text{tone_a}} = 440\text{Hz}$, $f_{\text{baseWiFi1}} = 2.4\text{Ghz}$, $f_{\text{baseWiFi2}} = 5\text{Ghz}$

Example: Sinusoidal Alternating Current

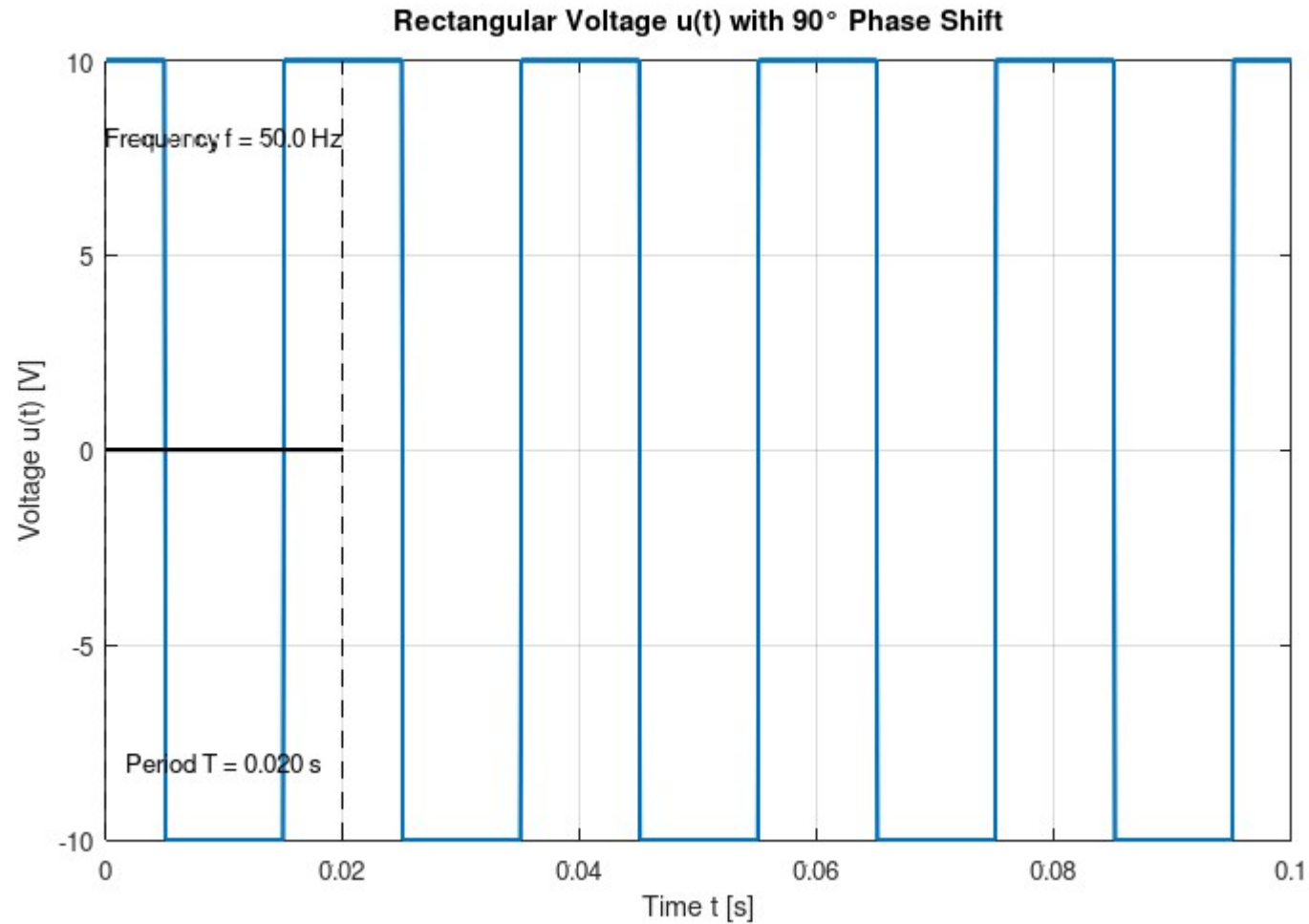


$$f = 1/T = 1/0,2s = 50Hz$$

Example: Rectangular Voltage



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Binary Data vs. Signal Transmission

- **Binary data are transferred digitally in computer networks (on higher layer)**
 - **Signal Transmission is always based on analogue signals (voltage, amplitude or light amplitude) on the transmission media (electrical, wireless or optical)**
 - **Signals are subject to physical laws**
 - Attenuation (all)
 - Reflections (electromagnetic waves, optical)
 - Shadowing (electromagnetic waves)
- physical limitations of bridgeable distances **without signal amplification**

Fourier Series (invented by Joseph Fourier, 1768 - 1830)



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- Any periodic function can be represented as a sum of sine and cosine waves
- The sine/cosine functions are multiple of a fundamental frequency
- The higher accuracy in approximation the higher frequencies are needed

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

where,

- $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$
- $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$
- $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$
- $n = 1, 2, 3, \dots$
- Ok, let's go into the beauty of Fourier Series (very detailed mathematics)
<https://youtu.be/wmCIrpLBFds>

Square Wave Approximation



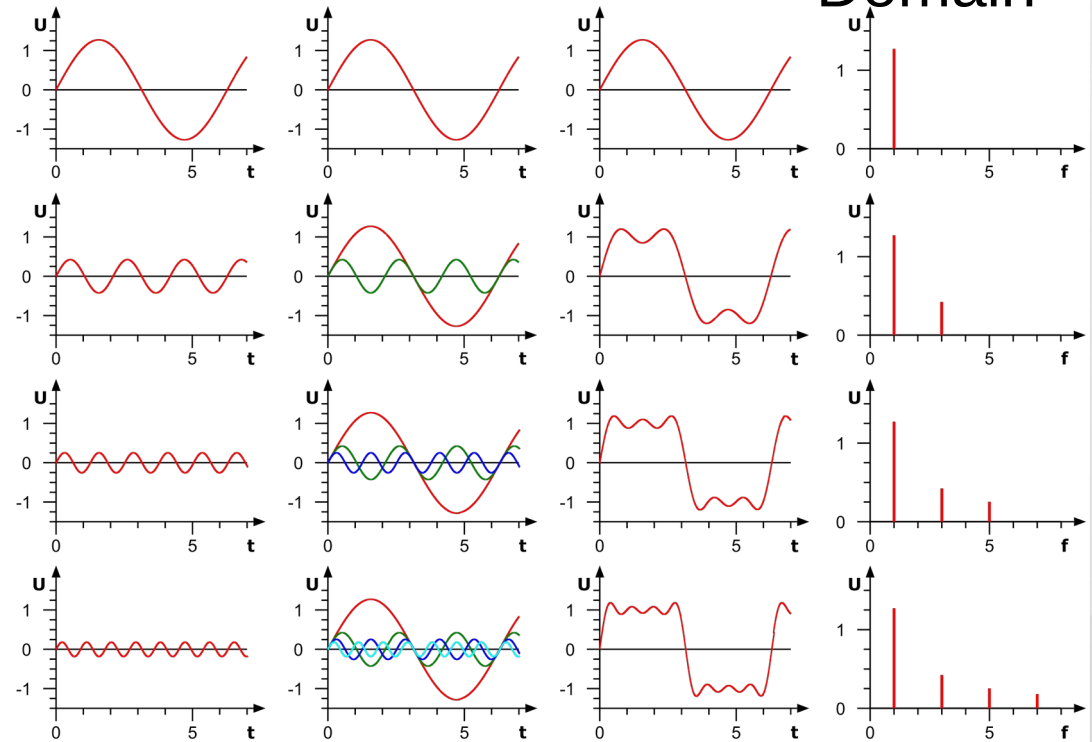
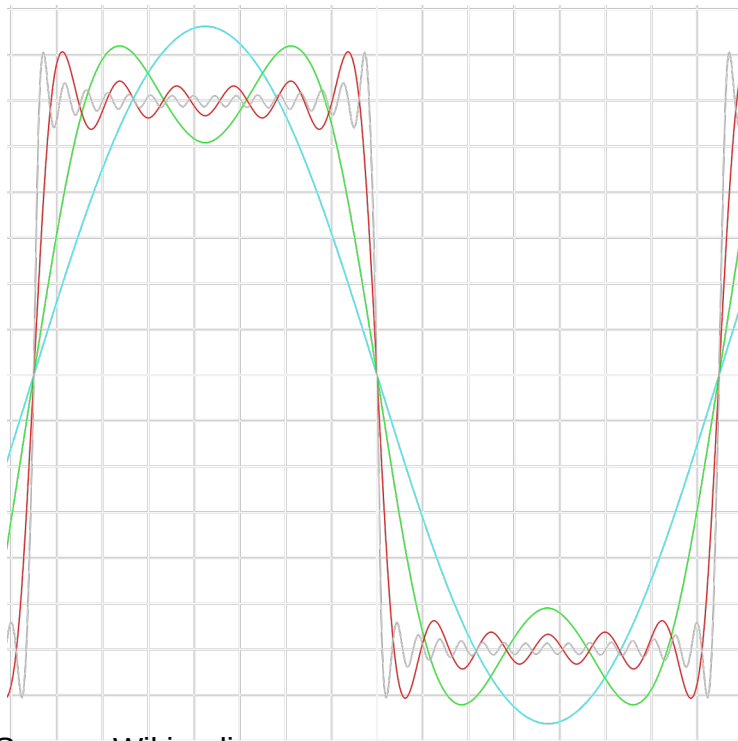
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A video on simulation of the approximation of square wave function

<https://youtu.be/k8FXF1KjzY0>

Time Domain

Frequency Domain



Source: Wikipedia.com

- **Generalization of Fourier Series leads to Fourier Transform**

Time Domain

Fourier
Transform

Frequency Domain

- **Mathematical method as basis for analogue + digital communication technologies and control theory**
 - Digital Signal Processing
 - Digital Filtering
 - i.e. MP3 compression/decompression
 -

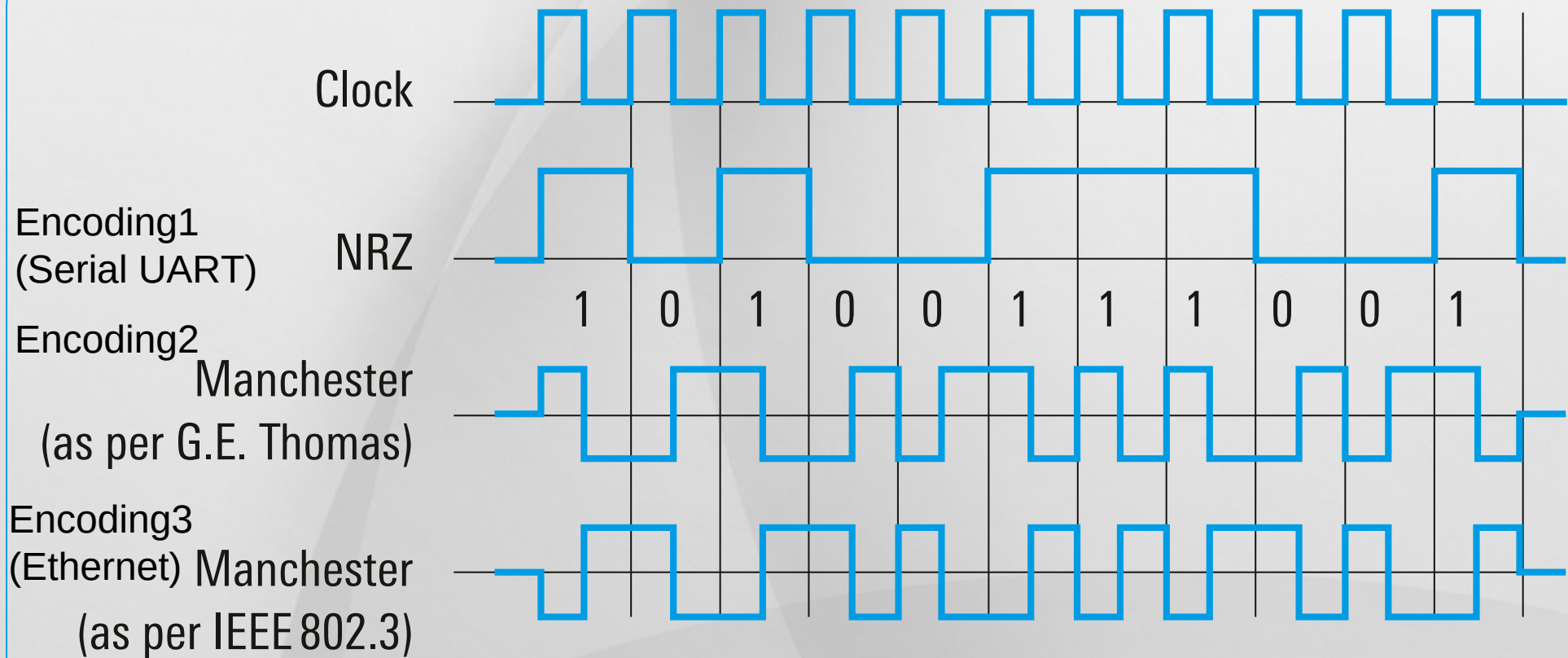
Fourier Series and Bandwidth

- **Bandwidth on a transmission media is the range of frequencies, which can be transmitted without interference**
- **Bandwidth = width of frequencies in the frequency domain**
- **Bandwidth limitations by attenuation of higher frequencies (acts like a low pass filter)**

Bit Rate vs Baud Rate

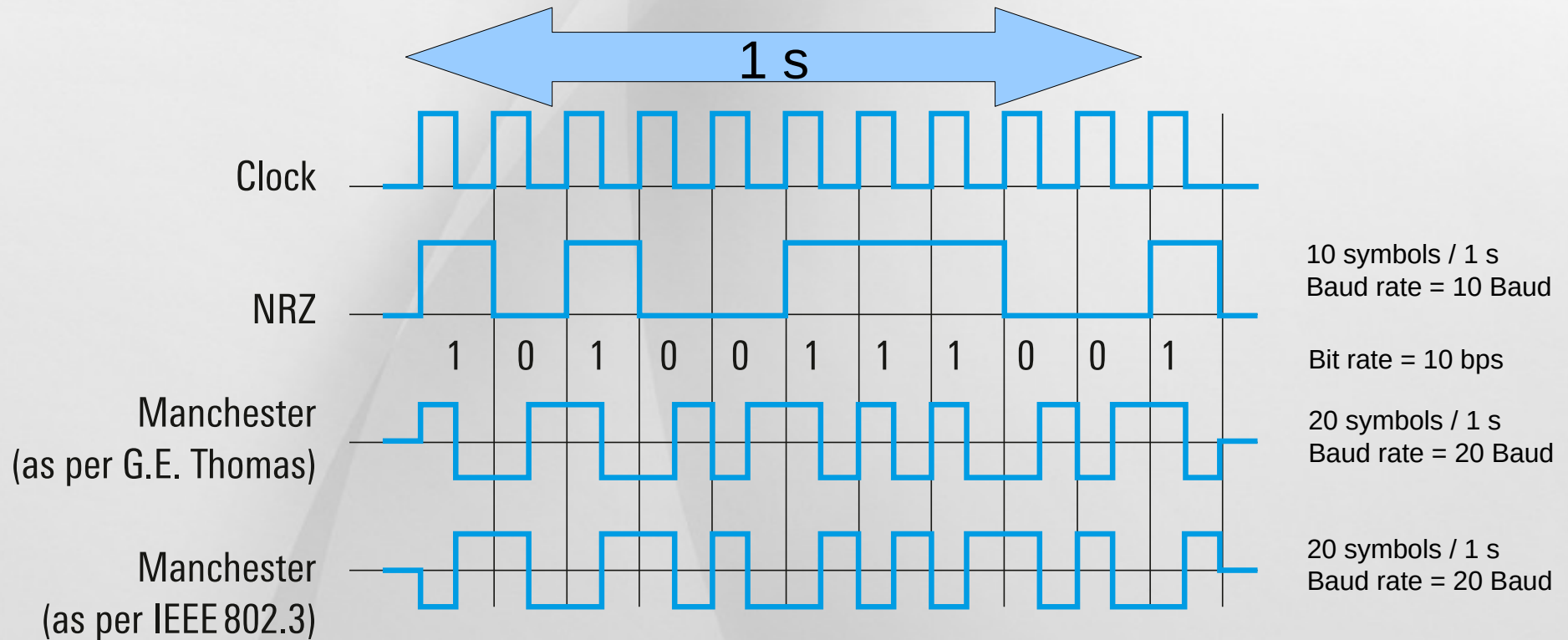
- **Bit rate: number of bits transferred per timespan**
 - Unit: **bits per second (bps), bit/s**
- **Baud rate: Number of symbols on the wire transferred per timespan**
 - Unit: symbols per second, baud
 - Initially: signaling rate of a telegraph = number of morse code characters per second
- **Ratio of bit rate and baud rate depends on the line encoding (Conversion of digital signals (in higher layers) ↔ digital signals on network media)**

Example: Bit Rate, Baud Rate, Encoding



Source: Rhode & Schwarz, https://cdn.rohde-schwarz.com/image/products/test-and-measurement/oscilloscopes/software/serial-triggering-and-decoding/manchester-and-nrz/manchester-coding-format-of-the-profibus-pa_standard-infographic-rohde-schwarz_200_114471_2.svg

Example: Bit Rate, Baud Rate Calculations



Source: Rhode & Schwarz, https://cdn.rohde-schwarz.com/image/products/test-and-measurement/oscilloscopes/software/serial-triggering-and-decoding/manchester-and-nrz/manchester-coding-format-of-the-profibus-pa_stand-infographic-rohde-schwarz_200_114471_2.svg

- **Bandwidth**
in the bit level transmission (above physical media, max. frequency spectrum)
 - Maximum number of bits being able to be transmitted per timespan
 - Aka **maximum bit rate**
 - Unit: bits per second
 - 1Mbit/s means „one million bits can be transmitted per second“, doubling would mean „two million bits can be transmitted per second“

- Timespan which is needed for a message to travel through a network from one end of the network to the most distant end
- Latency = Propagation delay + Transmission delay + Waiting time
- Propagation delay = (distance)/(speed of light c * velocity factor)
- $c = 299\,792\,458$ m/s (speed of light)

Media material	Vacuum	Twisted pair cable	Optical fiber
Velocity factor	1	0.6	0.67

- Transmission delay = message size / bandwidth
“Time needed for Transmission from the very first bit to last bit in the message“
- Waiting time caused by active network components (e.g. Switches etc.)

Bandwidth – Delay Product

Volume of a network connection

- The propagation speed is in any event limited by the speed of light and it depends on the velocity factor of the transmission medium
- The product of bandwidth and delay (latency) corresponds to the maximum number of bits that can reside inside the line between sender and receiver
- **Example:** A network with 100 Mbit/s bandwidth, and 10 ms latency
=> $100\,000\,000\text{ Bits/s} \times 0.01\text{ s} = 1\,000\,000\text{ Bits}$
- There are a maximum number of 1 000 000 Bits inside the network line, equivalent to 125 000 Bytes
= 125 kiB with $1\text{ kB} = 10^3\text{ Bytes}$
 $\approx 122\text{ kiB}$ with $1\text{ kiB (kibibyte)} = 2^{10}\text{ Bytes}$